

Spring Cloud Config Server

Objectifs :

1. Utiliser l'annotation @Value
2. Utiliser l'annotation @ConfigurationProperties
3. Spring Boot Profiles
4. Externaliser la configuration en utilisant JVM CLI
5. Spring Cloud Config Server
6. Lire la configuration à partir d'un dossier du classpath
7. Modifier department-microservice pour lire ses propriétés à partir de Config Server
8. Lire la configuration à partir du File system
9. Lire la configuration à partir du GITHUB
10. Actualiser la config avec l'option Refresh de actuator

Utiliser l'annotation @Value

Projet Department-microservice

application.yml

```
build:  
  version : 1.0
```

La classe DepartmentController

```
@Value("${build.version}")  
private String buildVersion;
```

```
@GetMapping("/version")  
public ResponseEntity<String> version()  
{  
    return ResponseEntity.status(HttpStatus.OK).body(buildVersion);  
}
```

Tester votre travail

localhost:8080/api/departments/version

Utiliser l'annotation @ConfigurationProperties

application.yml

```
author:
  name: "nadhém bel Hadj"
  email: "nadhemb@yahoo.com"
```

La classe Configuration

```
package com.nadhém.department.config;

import lombok.Data;
import org.springframework.boot.context.properties.ConfigurationProperties;
import org.springframework.stereotype.Component;

@Data
@Component
@ConfigurationProperties(prefix = "author")
public class Configuration {
    private String name;
    private String email;
}
```

La classe DepartmentController

```
@Autowired
Configuration configuration;
```

```
@GetMapping("/author")
public ResponseEntity<String> retrieveAuthorInfo() {
    return ResponseEntity.status(HttpStatus.OK)
        .body(configuration.getName() + " " + configuration.getEmail());
}
```

Spring Boot Profiles

Créer les fichiers suivants :

Le fichier application-prod.yml

```
spring:
  config:
    activate:
      on-profile: "prod"
author:
  name: "nadhém bel Hadj [PROD]"
  email: "nadhemb@yahoo.com [PROD]"
```

Le fichier application-qa. yml

```
spring:
  config:
    activate:
      on-profile: "qa"
author:
  name: "nadhém bel Hadj [QA]"
  email: "nadhemb@yahoo.com [QA]"
```

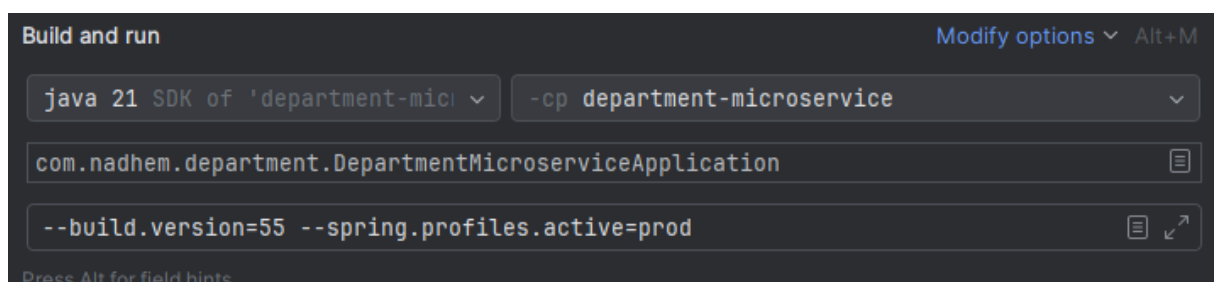
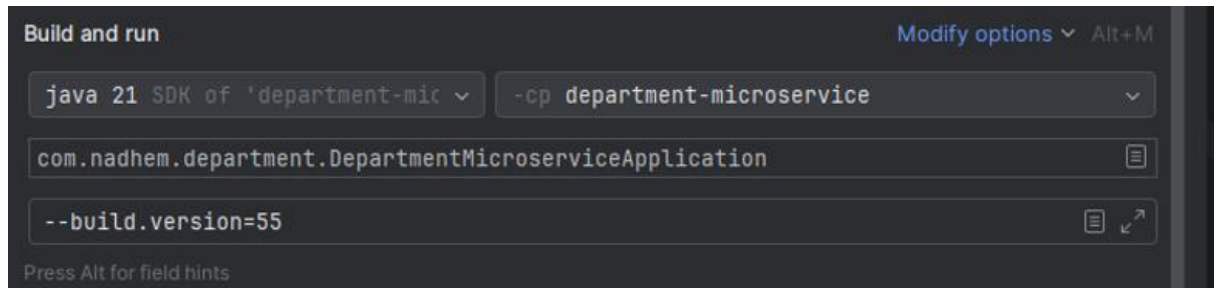
Le fichier application.yml

```
spring:
  config:
    import:
      - "application-prod.yml"
      - "application-qa.yml"
```

Activez-le profile « prod » comme profil de démarrage :

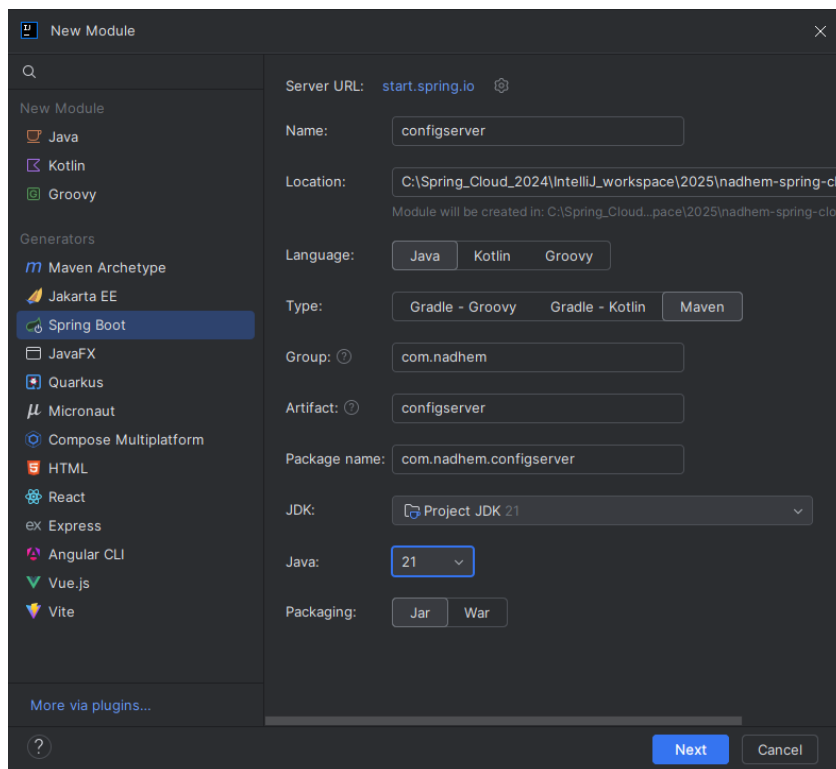
```
spring:
  profiles:
    active: "prod"
```

Externaliser la configuration en utilisant JVM CLI



Spring Cloud Config Server

Création du projet config Server



Added dependencies:

Config Server



Spring Boot Actuator

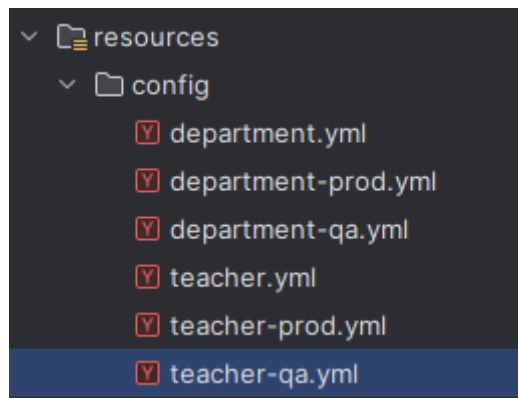


```
@SpringBootApplication
@EnableConfigServer
public class ConfigserverApplication {

    public static void main(String[] args) {
        SpringApplication.run(ConfigserverApplication.class, args);
    }

}
```

Lire la configuration à partir d'un dossier du classpath



depatment.yml

```
build:
  version: 1.0
author:
  name: "nadhém Department"
  email: "nadhemb_Department@yahoo.com"
```

depatment-prod.yml

```
build:
  version: 2.0
author:
  name: "nadhém [Department PROD]"
  email: "nadhemb_Department_PROD@yahoo.com"
```

depatment-qa.yml

```
build:
  version: 3.0
author:
  name: "nadhemi [ Department QA]"
  email: "nadhemi_Department_QA@yahoo.com"
```

teacher.yml

```
build:
  version: 1.0
author:
  name: "nadhemi teacher"
  email: "nadhemi_teacher@yahoo.com"
```

teacher-prod.yml

```
build:
  version: 2.0
author:
  name: "nadhemi [teacher PROD]"
  email: "nadhemi_teacher_PROD@yahoo.com"
```

teacher-qa.yml

```
build:
  version: 3.0
author:
  name: "nadhemi [ teacher QA]"
  email: "nadhemi_teacher_QA@yahoo.com"
```

Application.yml

```
spring:
  application:
    name: "configserver"
  profiles:
    active: native
  cloud:
    config:
      server:
        native:
          search-locations: "classpath:/config"
server:
  port: 9999
```

Modifier department-microservice pour lire ses propriétés à partir de Config Server

Modifier le nom de l'application :

```
spring:
  application:
    name: department
```

Le profile par défaut est « prod »

```
profiles:
  active: "prod"
```

Indiquez l'url du Config Server

```
config:
  import: "optional:configserver:http://localhost:9999/"
```

Enlevez ces lignes de application.yml du projet department-microservice car les pros build.version et author seront lues à partie du config server :

```
build:
  version: 1.0
author:
  name: "nadhém Department "
  email: "nadhemb_Department@yahoo.com"
```

Démarrer department-microservice, puis lancez :

The screenshot shows a REST client interface with the following details:

- Method:** GET
- URL:** localhost:8080/api/departments/author
- Params:** Authorization, Headers (7), Body, Pre-request Script, Tests, Settings
- Query Params:** A table with two rows, each containing a 'Key' column.
- Body:** Cookies, Headers (5), Test Results
- Response:** Pretty, Raw, Preview, Visualize, Text (selected), and a button with a red icon.
- Response Content:** 1 nadhem [Department PRODUCTION] nadhemb_Department_PROD@yahoo.com

Modifier le profile en utilisant « Modify Run Configuration... »

CLI arguments to your application: Alt+R

GET localhost:8080/api/departments/author

Params Authorization Headers (7) Body Pre-request Script Tests Settings

Query Params

Key
Key

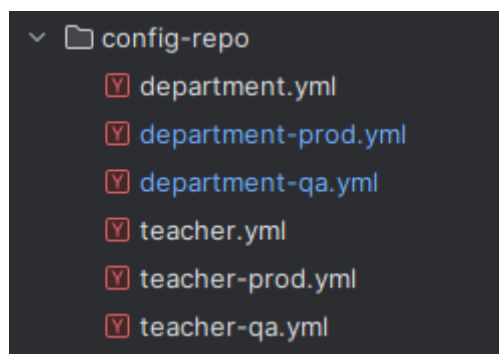
Body Cookies Headers (5) Test Results

Pretty Raw Preview Visualize Text

```
1  nadhem [ Department QA ] nadhemb_Department_QA@yahoo.com
```

Lire la configuration à partir du File system

Créer le dossier votre_projet/config-repo et y copiez les fichiers .yaml



Modifier le fichier application.yml :

```
config:
server:
  native:
    # search-locations: "classpath:/config"
    search-locations: file:///C:/Spring_Cloud_2024/IntelliJ_workspace/demo/nadhemb-spring-cloud/config-repo
```




Lire la configuration à partir du GITHUB

- Créer le repo « config-repo » sur Github
- Uploader les fichiers .yml du dossier config-repo vers le repo Github :

```
git init
git add .
git commit -m "first commit"
git branch -M main
git remote add origin https://github.com/nadhemBelHadj/config-repo.git
git push -u origin main
```
- Modifier le fichier application.yml

```
profiles:
# active: native
active: git
```

```
cloud:
config:
server:
# native:
# search-locations: "classpath:/config"
#search-locations: file:///C:/Spring_Cloud_2024/IntelliJ_workspace/2025/nadhem-spring-cloud/config-repo
git:
uri: https://github.com/nadhemBelHadj/config-repo.git
default-label: main
clone-on-start: true
```

localhost:9999/department/default

```
{
  "name": "department",
  "profiles": [
    "default"
  ],
  "label": null,
  "version": "489dbf8020fc4678d49fbfb3dbac91a65073cf29",
  "state": "",
  "propertySources": [
    {
      "name": "https://github.com/nadhemBelHadj/config-repo.git/department.yml",
      "source": {
        "build.version": 1,
        "author.name": "nadhem Department",
        "author.email": "nadhemb_Department@yahoo.com"
      }
    }
  ]
}
```

GET localhost:8080/api/departments/author

Params Authorization Headers (7) Body Pre-request Script Tests Settings

Query Params

	Key
	Key

Body Cookies Headers (5) Test Results

Pretty Raw Preview Visualize Text

1 nadhem Department on GIT HUB nadhemb_Department@yahoo.com

Actualiser la config avec l'option Refresh de actuator

Modifier la valeur dans le fichier department.yml au niveau de github :

author:

name: "nadhem Department on GIT HUB **Updated at 08:50**"

<http://localhost:8080/actuator>

```
"refresh": {  
  "href": "http://localhost:8080/actuator/refresh",  
  "templated": false  
},  
"config.client.version": "1.0.0",  
"author.name": "nadhem Department on GIT HUB Updated at 08:50"
```



http://localhost:8080/actuator/refresh

POST



http://localhost:8080/actuator/refresh

Params Authorization Headers (8) Body Pre-reqs

Query Params

	Key
	Key

Body Cookies Headers (5) Test Results

Pretty

Raw

Preview

Visualize

JSON



```
1 [
2   "config.client.version",
3   "author.name"
4 ]
```

 localhost:8080/api/departments/author

GET



localhost:8080/api/departments/author

Params

Authorization

Headers (7)

Body

Pre-request Script

Tests

Settings

Query Params

	Key
	Key

Body

Cookies

Headers (5)

Test Results

Pretty

Raw

Preview

Visualize

Text



1 nadhem Department on GIT HUB Updated at 09:00 nadhemb_Department@yahoo.com

```
..... ]  
}
```